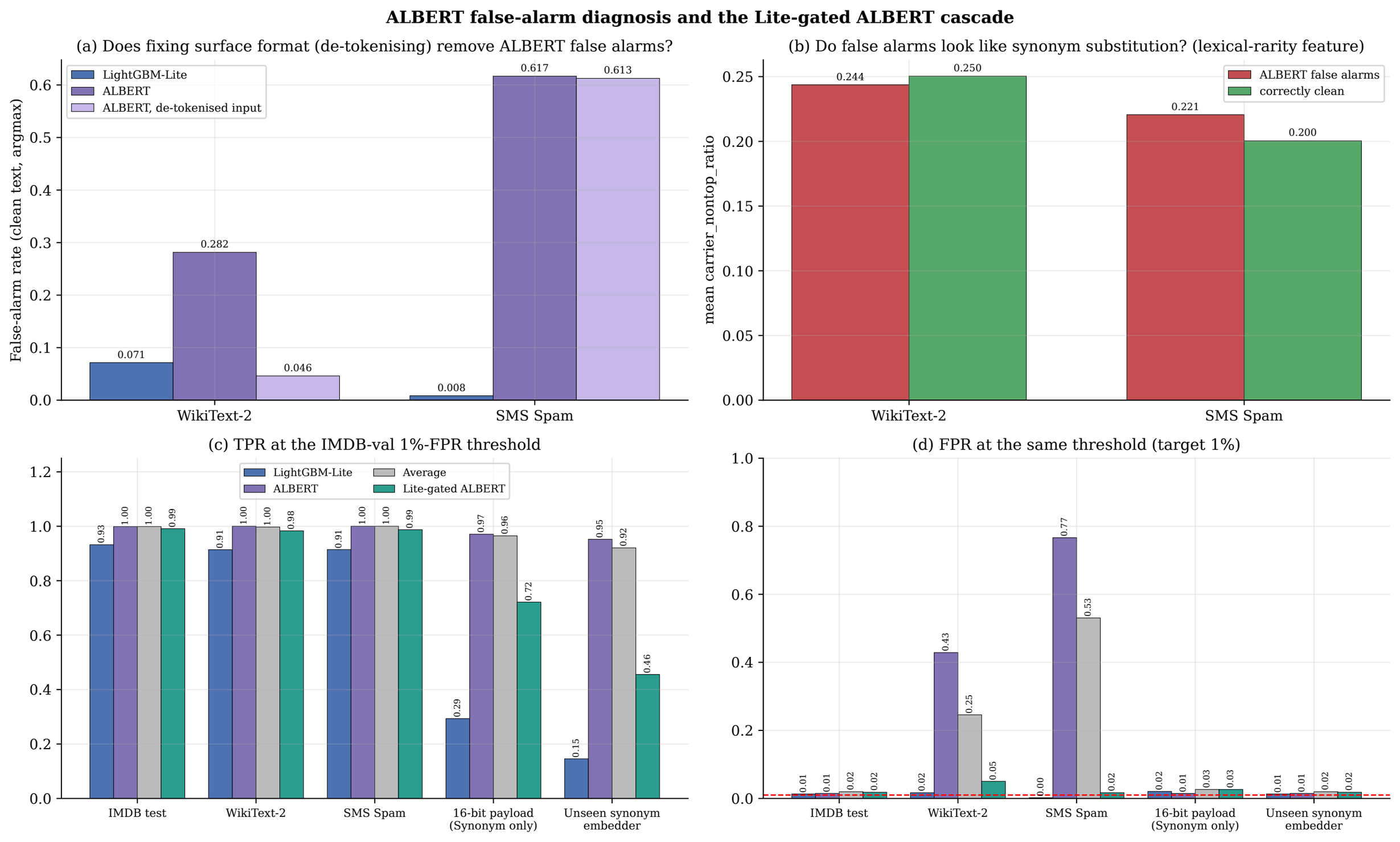
ALBERT false-alarm diagnosis and the Lite-gated ALBERT cascade
(a) Does fixing surface format (de-tokenising) remove ALBERT false alarms?
False-alarm rate (clean text, argmax)
LightGBM-Lite
ALBERT
ALBERT, de-tokenised input
0.071
0.282
0.046
0.008
0.617
0.613
WikiText-2
SMS Spam
(b) Do false alarms look like synonym substitution? (lexical-rarity feature)
mean carrier_nontop_ratio
ALBERT false alarms
correctly clean
0.244
0.250
0.221
0.200
WikiText-2
SMS Spam
(c) TPR at the IMDB-val 1%-FPR threshold
LightGBM-Lite
ALBERT
Average
Lite-gated ALBERT
0.93 1.00 1.00 0.99
0.91 1.00 1.00 0.98
0.91 1.00 1.00 0.99
0.29 0.97 0.96 0.72
0.15 0.95 0.92 0.46
IMDB test
WikiText-2
SMS Spam
16-bit payload (Synonym only)
Unseen synonym embedder
(d) FPR at the same threshold (target 1%)
0.01 0.01 0.02 0.02
0.02 0.43 0.25 0.05
0.00 0.77 0.53 0.02
0.02 0.01 0.03 0.03
0.01 0.01 0.02 0.02
IMDB test
WikiText-2
SMS Spam
16-bit payload (Synonym only)
Unseen synonym embedder